10 **(a)** State what is meant by redshift.

.....

.....

..... [2]

(b) Explain how observations of redshift lead to the idea that the universe is expanding.

.....

.....

..... [2]

(c) Explain how Hubble's law leads to the Big Bang theory of the origin of the universe.

.....

.....

.....

.....

..... [3]

[Total: 7]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN